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SUPERSEDING
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June 10, 1953

FEDERAL SPECIFICATION

ADAPTER, LUER NEEDLE, GLASS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a Luer type, glass, needle adapter (observation tube).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP–B–566—Boxes, Folding, Paperboard.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Box, Fiberboard.
PPP–B–676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 626—Glassware, Mehtods for Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL–L–10547—Liners, Case, Waterproof.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Standards Association, Inc. Standard:

Z70.1–1955—Glass and Metal Luer Tapers for Medical Applications, Dimensions of.

(Application for copies should be addressed to the American Standards Association, Incorporated, 10 East 40th Street, New York 16, N. Y.)

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3. REQUIREMENTS

3.1 Material. The needle adapters shall be fabricated of good quality clear borosilicate glass properly annealed and of such composition as will meet the requirements hereinafter specified. The glass shall have a coefficient of expansion not exceeding 0.0000055 per degree centigrade when tested as specified in 4.4.1.

3.2 Style, design, and dimensions. One end of needle adapter (observation tube) shall have a ground glass Luer taper connection and the other end shall fit rubber tubing. The design of the adapter shall be as shown on figure 1 of this specification and in accordance with figure 1 of ASA Standard Z70.1–1955 for Luer tip.

3.3 Annealing. The fabricated needle adapter shall be free of ring strain when tested as specified in 4.4.2. A slight degree of longitudinal strain is permissible.

3.4 Corrosion resistance. When submitted to the boiling test, as specified in 4.4.3, the needle adapters shall show no evidence of corrosion, scumming, chipping, or cracking.

3.5 Solubility. When tested as specified in 4.4.4, the solubility of the glass shall be such as to require not more than 1.0 ml. of the 0.020N sulfuric acid to neutralize the extract from 10 gm. of glass.

3.6 Thermal shock resistance. When submitted to the shock test, as specified in 4.4.5, the needle adapters shall not crack, chip, or break.

3.7 Male Luer slip tip. The male Luer slip tip shall make secure, watertight union with a needle hub having a corresponding taper. The tip shall be tested for proper taper and dimensions as specified in 4.4.6. When so tested, tips shall fit the ring gage within the "GO"-"NO GO" limits of the gage as shown on figure 7 of ASA Standard Z70.1–1955. Tips that do not fit the gage within these limits shall be considered not acceptable. In addition, when this ring gage is

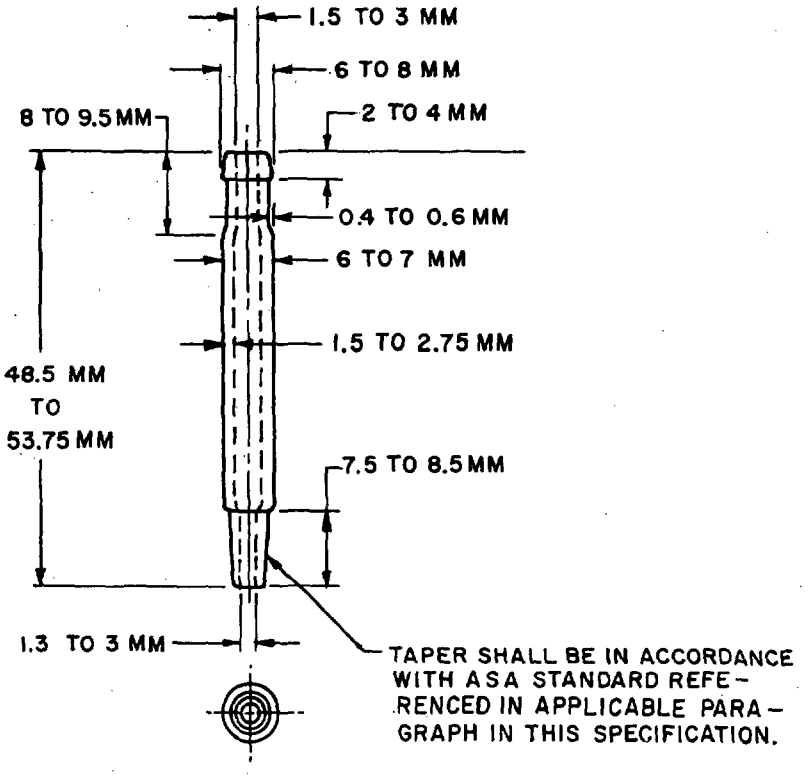


FIG. 1

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fitted onto the tip as specified above, there shall be no movement of the gage on the tip when the gage is pressed lightly from side to side in all directions. The tips shall also be tested for fluid tightness as specified in 4.4.7 and when so tested shall not show leakage exceeding the formation of a falling drop within 10 seconds.

3.8 Workmanship. Workmanship shall be first class throughout. Adapters shall be smoothly and symmetrically finished. Glass shall be uniformly distributed, properly annealed, and shall be free from seeds, blisters, and other defects or irregularities which detract from its appearance or which may impair its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Inspection.* Inspection as used in this specification is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* The inspection lot shall be determined by the cognizant government inspector and shall be based on the unit of product being one tube (end unit).

4.2.2 *Sampling for examination.* A random sample of tubes shall be selected from each lot offered to the Government in accordance with Military Standard MIL–STD–105. Examination for dimensions shall be in accordance with Military Standard MIL–STD–105 and as indicated in table I.

4.2.3 *For tests.* Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105 and as indicated in table I.

TABLE I.—*Sampling criteria*

Component	Characteristic	Req.	Test procedure	Rqmt. Appl.		Unit of product	Insp. level	AQL
				Indiv. unit	Lot avg.			
Adapter	Coefficient of expansion	3.1	4.4.1		X	*	*	*
	Dimensions	3.2	....	X		1 adapter	L3	4.0
	Annealing	3.3	4.4.2	X		1 adapter	L3**	***
	Corrosion resistance	3.4	4.4.3	X		1 adapter	L3**	***
	Solubility	3.5	4.4.4		X	*	*	*
	Thermal shock resistance	3.6	4.4.5	X		1 adapter	L3**	***
	Taper	3.7	4.4.6	X		1 adapter	II	1.0
	Fluid tightness	3.7	4.4.7	X		1 adapter	L7	1.5

* Sampling shall be as indicated in applicable test method. Each lot of glass shall be tested.
** A minimum of three units shall be tested regardless of lot size.
*** Acceptance number shall be zero for all sample sizes.

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4.2.4 Where possible, the same unit of product shall be used for the determination of two or more test characteristics.

4.3 Examination. The adapter shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests.

4.4.1 *Thermal coefficient of expansion.* The thermal expansion shall be determined in accordance with either method 240 or 241 of Federal Test Method Standard No. 626.

4.4.2 *Annealing test.* The specimens shall be examined for ring strain and longitudinal strain in accordance with method 420 of Federal Test Method Standard No. 626.

4.4.3 *Corrosion test.* Fifty percent of the adapters taken for test shall be completely immersed in distilled water in a closed vessel having a small steam vent and boiled continuously for 6 hours, boiling water being added to compensate for evaporation losses. The adapters shall be suspended in the water so that they do not contact the containing vessel.

4.4.4 *Solubility test.* The solubility shall be determined in accordance with method 320 of Federal Test Method Standard No. 626.

4.4.5 *Thermal shock test.* The shock test shall be performed in accordance with method 401 of Federal Test Method Standard No. 626. The oven shall be maintained at a temperature of 100°C.

4.4.6 *Taper of male Luer slip tip.* The male Luer tip shall be tested with the ring gage shown on figure 7 of ASA Standard Z70.1–1955. This gage shall be slipped onto the tip with very light pressure.

4.4.7 *Fluid tightness for male Luer slip tips.* A gage, as shown on figure 8 of the ASA Standard Z70.1–1955 (except that it shall be constructed with a suitable fluid path), shall be attached to the male Luer slip. The free end of the adapter (tubing end) shall be plugged and the water under pressure of 45 pounds per square inch (p.s.i.) shall be let in through the gage.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging and packing used are defined in Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.* Twelve adapters shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.1.1 *Intermediate package.* Twelve packages as specified in 5.1.1 shall be overpacked in a carton conforming to Federal Specification PPP–B–636, type I, class 1, style RSC. Each carton shall be adequately secured to prevent accidental opening.

5.1.2 *Level C.* Adapters shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.* Twenty-four intermediate containers as specified in 5.1.2 shall be packed in exterior containers designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–621, table III. Each box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Strapping, when required, shall be in accordance with applicable box specification.

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NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.* Twenty-four intermediate containers as specified in 5.1.2 shall be packed in exterior containers constructed in accordance with Federal Specification PPP–B–621, table I, or PPP–B–636, type I class 1, fabricated from not less than 275-pound test fiberboard.

5.2.3 *Level C.* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, unit, intermediate, and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit, intermediate, and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Glassware, chemical laboratory.

Carload minimum weight 18,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Glassware, laboratory.

Motor volume minimum weight 18,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnshed, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—M
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Copies of this specification may be purchased for 5 cents each.

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